



Rice Price Prediction in Sri Lanka

A basic Python solution for forecasting rice prices using economic indicators

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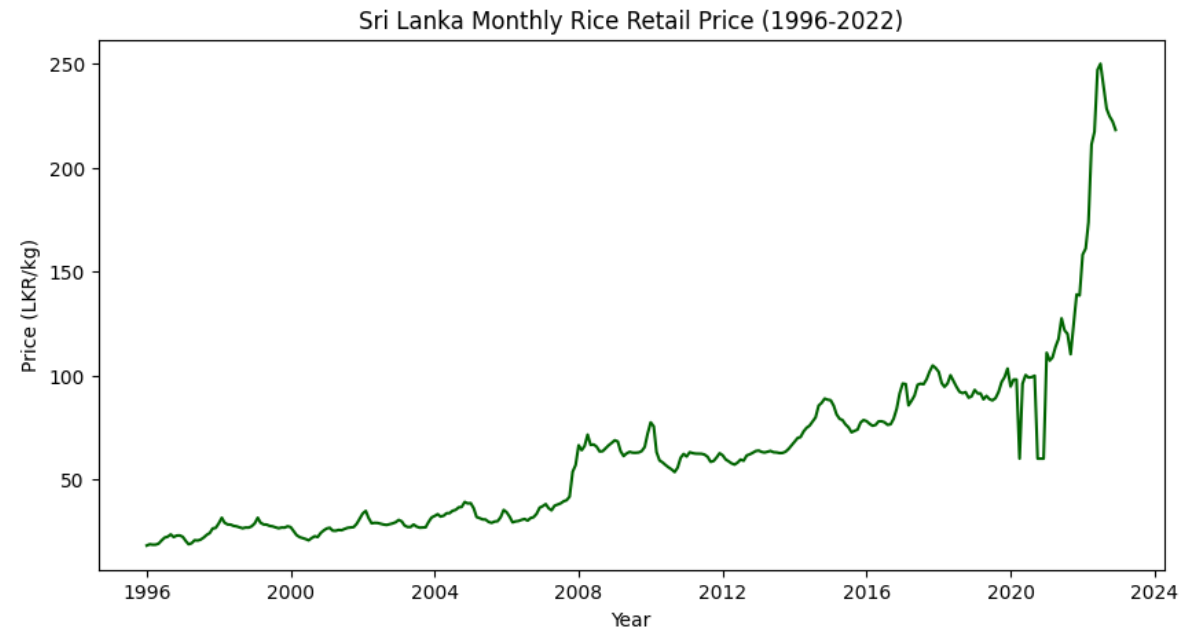


The Problem

- Rice is Sri Lanka's staple food — and its price has become highly unstable.
- During the 2021–2022 economic crisis, retail prices surged rapidly.
- Sudden spikes hit low-income households hardest — a direct food-security issue.
- **Question: can we predict rice price from related economic indicators?**

2.6×

Price increase





The Dataset

Source

SriOryzia: Multivariate Rice Price Forecasting — Kaggle (luqmanrumaiz) [Dataset Link](#)

Coverage

324 monthly records, Jan 1996 – Dec 2022

Key columns

retail price, producer prices (3 districts), exchange rate, fuel price, money supply (M0/M1/M2/M2B)

Snapshot

date	price	Exchange rate
1996-01	18.07	53.69
2020-01	94.66	181.9
2022-07	250.07	360.9

- Missing values in price & producer-price columns — handled during cleaning.
- Dataset is real, published (not synthetic).



Proposed Solution

A Linear Regression model that predicts monthly retail rice price from related economic indicators

Inputs (features)

Producer prices (3 districts), exchange rate, fuel price, money supply (M0/M1/M2/M2B)

Model

Linear Regression (scikit-learn) — 80/20 train-test split

Output

Predicted monthly rice retail price (LKR/kg)

Python Implementation

1. Clean — fill missing values

```
def fill_missing_with_median(data,
    columns):
    for col in columns:
        if data[col].isnull().sum() > 0:
            median = data[col].median()
            data[col] =
data[col].fillna(median)
    return data
```

2. Train the model

```
X_train, X_test, y_train, y_test = \
    train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)
```

Python concepts demonstrated

Functions

```
load_data(),
fill_missing_with_median()
```

Exception handling

```
try/except around file loading
```

Libraries

```
pandas, matplotlib, scikit-learn
```



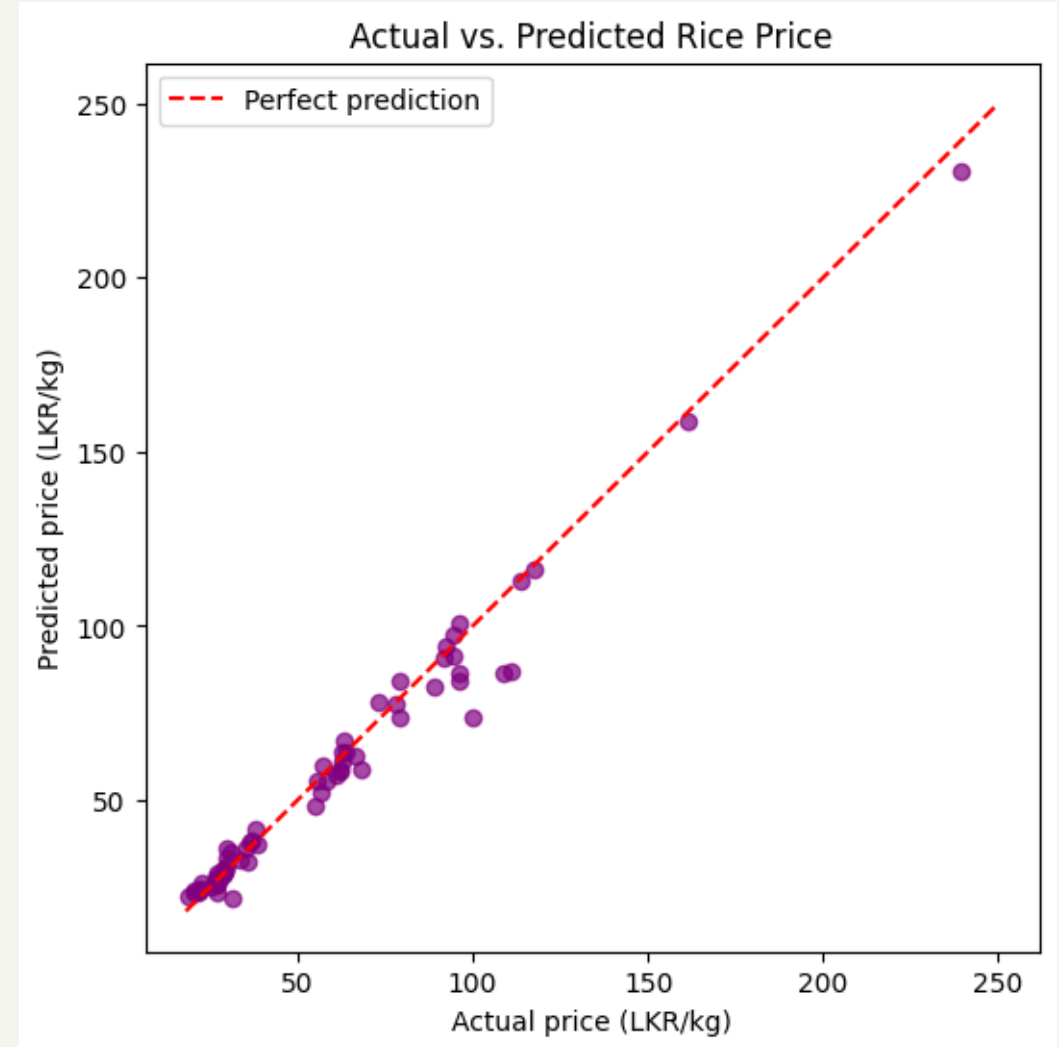
Results

$$R^2 = 0.97$$

of price variation explained by
the model

$$MAE \approx 4 \text{ LKR/kg}$$

average prediction error





Conclusion

A simple Linear Regression model can predict Sri Lankan rice prices with high accuracy from producer prices, exchange rate, fuel price, and money supply.

Limitations

- Only 324 monthly records — a small dataset
- Linear model may not fully capture the 2022 shock
- Data stops at December 2022

Future Improvements

- Try non-linear models (e.g. Random Forest)
- Add more recent data (2023–2026)
- Include rainfall data — affects paddy production